Statement of purpose

My AI/ML Journey: Exploration, Evolution, and Intent

My technical journey in AI/ML stems from a desire to turn raw data into predictive intelligence. As a Computer Science undergraduate maintaining a 9.65 CGPA, I began by mastering the foundational mathematical and data-driven prerequisites. I immersed myself in data preprocessing, exploratory data analysis (EDA), and model evaluation using Python's core ecosystem: NumPy, Pandas, Scikit-Learn, Matplotlib, and Seaborn.

To ground this theory, I built practical pipelines targeting real-world problems. I developed a *Predictive Housing Analytics Engine*, implementing robust pipelines to handle missing values, eliminate outliers, and apply one-hot encoding before optimizing Linear Regression and Random Forest models. Moving from regression to classification, I engineered an end-to-end *Customer Churn Prediction Pipeline*. This project forced me to tackle real-world data complexities, such as managing class imbalances through sampling methodologies and applying feature scaling. I then deployed it using a lightweight Flask API to expose real-time churn scores. Furthermore, my current role as a Software Development Intern and my projects with the MERN stack have taught me how to integrate machine learning components into scalable, production-ready applications.

While I have built a reliable foundation in classical machine learning, significant gaps remain between my academic exploration and industry-scale deployment. I lack deep, structured exposure to advanced Deep Learning architectures, Large Language Models (LLMs), and geometric deep learning. Crucially, while I can train models locally, I need to bridge the gap in understanding how to scale, optimize, and deploy these models to handle enterprise-level data velocity and volume. The Amazon ML Summer School (MLSS) offers the precise, rigorous curriculum and mentorship from elite scientists I need to transform my localized knowledge into production-grade competency.

My unique combination of intense curiosity, project execution, and algorithmic grit makes me an ideal fit for MLSS. My curiosity is driven by problem-solving; I have solved over 340+ LeetCode problems and achieved a 3-Star CodeChef rating, which trains me to optimize code under strict constraints—a vital skill for resource-efficient ML. Having proved my ability to collaborate under pressure by ranking in the Top 45 out of 422 teams in the SIH Internal Hackathon, I am eager to bring this competitive, collaborative energy to the MLSS cohort and master the frontier of machine learning.